

11 PFRs

Why are we learning this?

In the previous lecture, we learned how to design Continuous Stirred Tank Reactors (CSTRs). Today we will study another important reactor type: the **Plug Flow Reactor (PFR)**. Unlike a CSTR, a PFR is not perfectly mixed: Conditions change continuously as material travels through the reactor. This difference has major consequences for reactor performance and reactor volume.

By the end of today's lesson, you should understand:

- The physical meaning of the PFR mole balance.
- Why conditions vary throughout a PFR.
- Why PFR design equations are differential equations.
- Why PFRs are often more volume-efficient than CSTRs.
- How to apply the PFR mole balance to reactor design problems.
- How reactor sequencing influences conversion.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



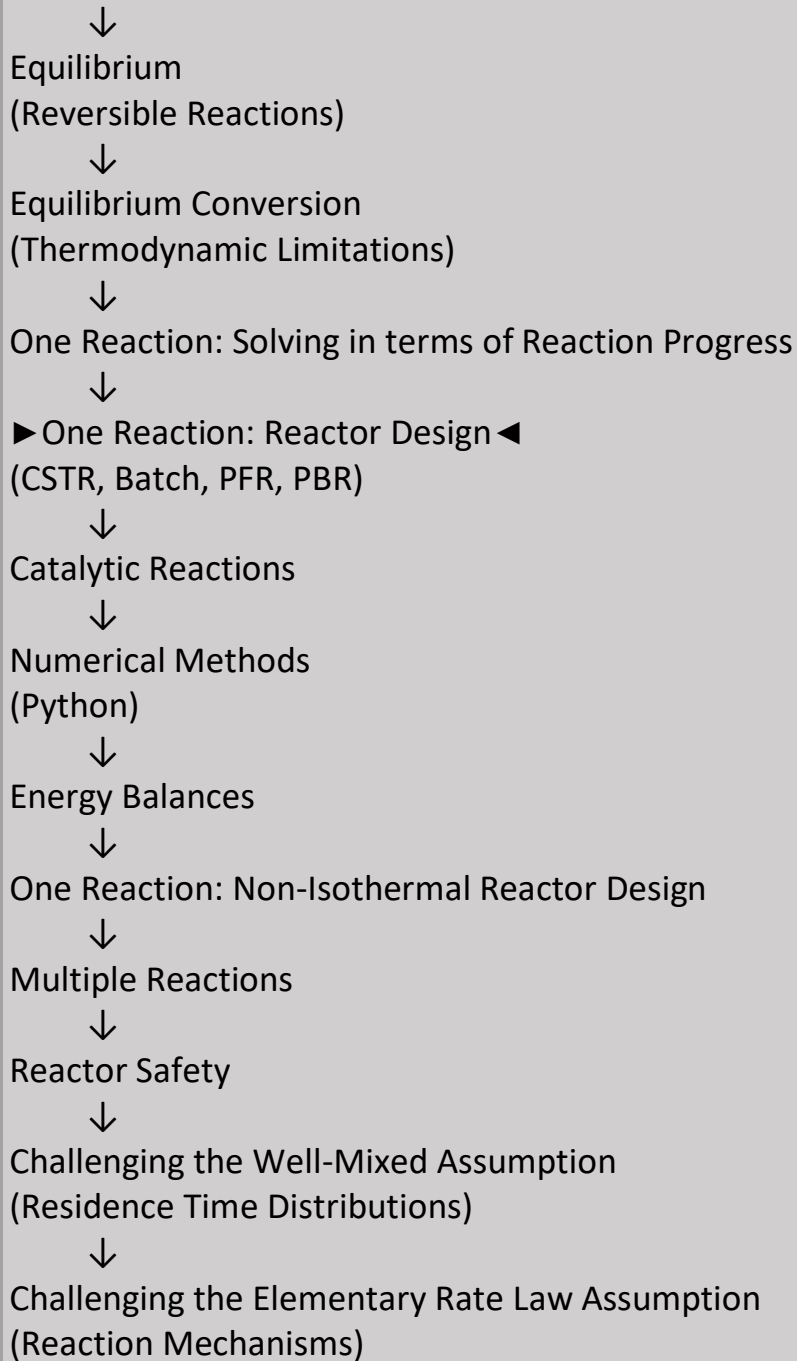
Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Rate Laws
(Kinetics)



Big Idea

A PFR design problem usually asks: How much reactor volume is needed to achieve a desired conversion? Unlike a CSTR, the reaction rate changes continuously throughout the reactor

PFR mole balance:

- In terms of number of moles (F): $\frac{dF_A}{dV} = r_A$. In words, the rate of change in the molar flowrate of species A is equal to the rate of generation of A.
- If species A is a reactant, the rate of generation of A is
- If species A is a product, the rate of generation of A is

What Makes a PFR Different?

A PFR is not perfectly mixed. As material flows through the reactor:

- Concentrations change.
- Conversion changes.
- Reaction rate changes.

Different locations within the reactor operate under different conditions. This is the defining feature of a PFR.

What Does the PFR Mole Balance Mean?

The PFR balance describes how the molar flow rate changes as fluid moves through reactor volume. The equation describes how reaction progresses continuously throughout the reactor.

Why Is the PFR Balance a Differential Equation?

Conditions inside a PFR are not uniform. Every axial location in the reactor has:

- a different concentration,

- a different conversion,
- a different reaction rate.

Because conditions vary continuously, the mole balance must describe changes that occur over very small portions of reactor volume.

That is why the balance appears in differential form.

- In terms of conversion (X): $\frac{dX_A}{dV} = \frac{-r_A}{F_{A0}}$
 - Notice that this balance is written with respect to the rate of consumption of A, because

Why Use Conversion?

For one, it simplifies the problem into "one equation (mole balance) and one unknown (conversion)". Further, it's a direct indication of reactor performance. Rather than asking:

How much A remains?

we can ask:

What conversion was achieved?

Writing the balance in terms of conversion makes design calculations easier to interpret and compare across operating conditions.

Why Are PFRs Often More Efficient?

In a PFR: The inlet section operates at high concentration.

Higher concentration usually means: Higher reaction rate.

As a result, most of the reactor operates at a faster rate than the outlet.

A CSTR cannot take advantage of this because the entire reactor operates at outlet conditions.

This is why PFRs can require less volume than CSTRs to achieve the same conversion.

- Let's do an example: The reversible reaction $A \rightleftharpoons B$ is carried out isothermally and isobarically in a PFR. The equilibrium constant at the reaction temperature is 5.8. The overall conversion is 0.55. A second PFR that is identical in every way to the first is placed downstream of the first PFR, so that the two PFRs operate in series. Determine the overall conversion for the system. Pure A is fed to the system in both cases.

Use the Established Design Equation

For standard PFR problems, the reactor design equation is already known. You do not need to re-derive it unless:

- the problem explicitly requests a derivation, or
- the reactor does not satisfy PFR assumptions.

Once the governing equation has been established, reactor design focuses on applying it correctly.

Think Before Calculating

Before starting a PFR design calculation, identify:

- What quantity is being solved for?

- What rate law applies?
- How does concentration depend on conversion?
- Is equilibrium important?

Building the physical picture first often makes the mathematics much easier.

Why Put Reactors in Series?

A PFR continues reacting as long as:

- reactants remain, and
- equilibrium has not been reached.

Placing reactors in series increases total reactor volume. Additional volume allows the system to move closer to equilibrium conversion.

This idea becomes important when designing industrial reactor systems.

Summary and Key Takeaways

Today we learned how to apply the Plug Flow Reactor (PFR) design equation.

Key Ideas

- ✓ A PFR is not perfectly mixed. Conditions change continuously throughout the reactor.
- ✓ Concentration, conversion, and reaction rate vary with axial position.
- ✓ The PFR mole balance therefore appears as a differential equation.
- ✓ Conversion is the most useful variable for reactor design calculations involving a single reaction.
- ✓ PFRs typically use reactor volume more efficiently than CSTRs because portions of the reactor operate at higher concentrations and hence faster reaction rates.
- ✓ Reactor sequencing can be used to increase overall conversion and move the system closer to equilibrium.

Most Important Idea

PFR: Conditions change throughout the reactor.

CSTR: Conditions are the same everywhere.

This single difference explains why the design equations are different and why PFRs and CSTRs often achieve different conversions for the same reactor volume.